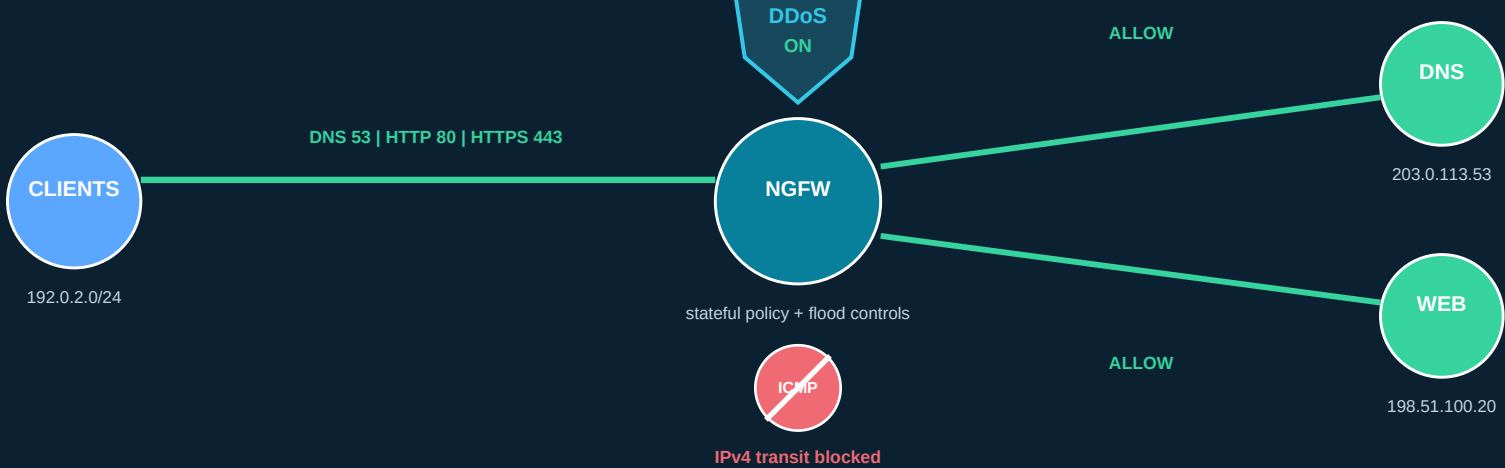


Firewall Policy + DDoS Defense Playbook

Allow DNS, HTTP, and HTTPS | Block IPv4 ICMP and unmatched transit | Enable SYN, UDP, and ICMP flood protection

PUBLIC-SAFE FIREWALL LAB

Reserved example networks | no production identifiers | illustrative thresholds



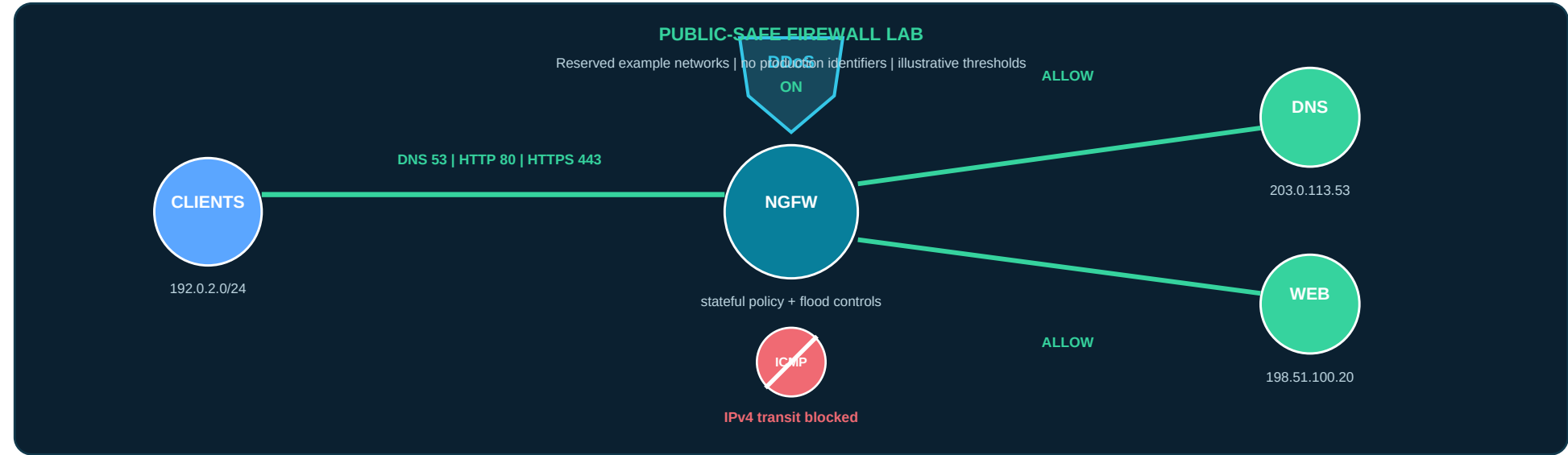
10 FIREWALL PLATFORMS	4 ALLOWED TRANSPORTS	IPv4 ICMP BLOCKED	DDOS CONTROLS ENABLED
CLI and manager-driven workflows	TCP/53 UDP/53 TCP/80 TCP/443	Transit policy scope	SYN UDP ICMP flood defense

Prepared by Marc Lavoie | Technical portfolio reference | August 2026

01 | FIREWALL ENGINEERING PLAYBOOK

One simple policy, applied consistently

This lab demonstrates disciplined firewall reasoning without pretending that every platform is configured the same way. The intent remains constant: narrowly allow required web and DNS traffic, deny IPv4 ICMP transit, deny everything else in scope, and turn on measurable flood protection.



Order	Rule	Source	Destination	Service	Action	Logging
10	ALLOW-DNS	LAB-CLIENTS	LAB-DNS	TCP/53 + UDP/53	Allow	Session end
20	ALLOW-WEB	LAB-CLIENTS	LAB-WEB	TCP/80 + TCP/443	Allow	Session end
30	DENY-ICMPV4	LAB-CLIENTS	LAB-UNTRUST	IPv4 ICMP	Drop	Every match
40	DENY-OTHER	LAB-CLIENTS	LAB-UNTRUST	Any	Drop	Rate-limited

FAKE LAB SOURCES

LAB-CLIENTS = 192.0.2.0/24. LAB-DNS = 203.0.113.53. LAB-WEB = 198.51.100.20. All are documentation-only addresses.

STATEFUL RETURN

Allowed sessions use the firewall state table for return traffic. No broad reverse-direction allow rule is added.

ICMP SCOPE

The exercise blocks IPv4 ICMP transit. It does not disable firewall management-plane services or essential ICMPv6 behavior.

WHY ICMPV6 IS DIFFERENT

Do not copy the IPv4 deny into an IPv6 policy as a blanket ICMPv6 block. Neighbor Discovery, Packet Too Big, and other ICMPv6 functions are required for healthy IPv6 operation.

02 | FIREWALL ENGINEERING PLAYBOOK

DDoS controls are enabled, measured, and verified

The activation commands below use intentionally fake lab thresholds. They demonstrate the mechanism; they are not production recommendations. Real thresholds must come from observed peak traffic, platform capacity, application requirements, and upstream circuit limits.

Flood type	Illustrative lab threshold	Protection behavior	Evidence
TCP SYN	Alarm 1,200 CPS; activate 2,000; max 4,000	Alarm, SYN cookies/proxy, RED, rate limit, or drop depending on platform.	Protection state, counters, half-open sessions, CPU.
UDP	Alarm 5,000 PPS; activate 8,000; max 10,000	Rate-limit or drop excess packets to a destination or ingress zone.	UDP flood counters, drops, permitted DNS health.
ICMP	Alarm 150 PPS; activate 300; max 500	Drop or rate-limit flood traffic even though policy already denies IPv4 ICMP transit.	ICMP flood counter and deny-rule matches.
Connections	1,500 new connections per second	Per-server or per-client quotas where the platform supports them.	Connection rate, quota drops, application health.

Activation sequence

1	Baseline	Record normal and peak CPS/PPS, concurrent sessions, CPU, and interface utilization.
2	Monitor	Apply alarm or monitor-only settings first where supported; confirm expected logs and counters.
3	Protect	Enable SYN, UDP, ICMP, and connection-rate controls with rollback access available.
4	Exercise	Generate bounded authorized test traffic and confirm legitimate DNS/web sessions remain healthy.
5	Tune	Raise or lower thresholds from evidence; preserve an upstream mitigation plan for circuit saturation.

ON-BOX VALUE Protects sessions, state tables, CPUs, and published services after attack traffic reaches the firewall.	UPSTREAM LIMIT Cannot recover bandwidth once a volumetric attack saturates the ISP circuit. Carrier filtering, RTBH, FlowSpec, or scrubbing may still be required.	TEST BOUNDARY Use an isolated lab or authorized test window. Never direct flood testing at public systems or networks you do not own.
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Platform commands: PAN-OS and FortiOS

Palo Alto Networks PAN-OS CLI ZONE PROTECTION Security policy blocks IPv4 ICMP transit; Zone Protection is attached to the ingress untrust zone. Block IPv4 ICMP transit <pre>configure set rulebase security rules DENY-ICMPV4 from LAB-TRUST to LAB-UNTRUST set rulebase security rules DENY-ICMPV4 source any destination any set rulebase security rules DENY-ICMPV4 application ping set rulebase security rules DENY-ICMPV4 service application-default set rulebase security rules DENY-ICMPV4 action deny log-end yes</pre> Enable flood / DDoS protection <pre>set network profiles zone-protection-profile LAB-EDGE-ZP flood tcp-syn enable yes set network profiles zone-protection-profile LAB-EDGE-ZP flood tcp-syn red alarm-rate 12 set network profiles zone-protection-profile LAB-EDGE-ZP flood tcp-syn red activate-rate 12 set network profiles zone-protection-profile LAB-EDGE-ZP flood tcp-syn red maximal-rate 12 set network profiles zone-protection-profile LAB-EDGE-ZP flood udp enable yes set network profiles zone-protection-profile LAB-EDGE-ZP flood udp red activate-rate 100 set network profiles zone-protection-profile LAB-EDGE-ZP flood icmp enable yes set network profiles zone-protection-profile LAB-EDGE-ZP flood icmp red activate-rate 36 set zone LAB-UNTRUST network zone-protection-profile LAB-EDGE-ZP commit</pre> Verify <pre>show config running match LAB-EDGE-ZP show counter global filter severity drop</pre>	Fortinet FortiGate FORTIOS CLI DOS POLICY An explicit ICMP deny is logged before the cleanup rule. The DoS policy is evaluated for matching ingress traffic. Block IPv4 ICMP transit <pre>config firewall policy edit 30 set name "DENY-ICMPV4" set srcintf "lan" set dstintf "wan1" set srcaddr "all" set dstaddr "all" set action deny set service "PING" set logtraffic all next end</pre> Enable flood / DDoS protection <pre>config firewall DoS-policy edit 0 set name "LAB-EDGE-DOS" set interface "wan1" set srcaddr "all" set dstaddr "all" set service "ALL" config anomaly edit "tcp_syn_flood" set status enable set log enable set action block set threshold 2000 next edit "udp_flood" set status enable set action block set threshold 8000 next edit "icmp_flood" set status enable set action block set threshold 300 next end next end</pre> Verify <pre>show firewall policy show firewall DoS-policy diagnose sys session list</pre>
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Platform workflows: Cisco and Check Point

Cisco Secure Firewall

FMC / FDM / CDFMC | MANAGER-DRIVEN

FTD persistent access control is deployed by its manager. The honest command is a management workflow, not an ASA ACL pasted into diagnostic CLI.

Block IPv4 ICMP transit

```
FMC > Policies > Access Control
1. Add rule: DENY-ICMPV4
2. Zones: LAB-TRUST -> LAB-UNTRUST
3. Protocol: ICMPV4
4. Action: Block with logging
5. Place above default block
6. Deploy to the assigned FTD
```

Enable flood / DDoS protection

```
FMC > Policies > Access Control > Advanced Settings
1. Open Threat Defense Service Policy
2. Add external/server traffic class
3. Enable TCP Intercept / embryonic limits
4. Add measured connection limits
5. Save, deploy, and monitor drops

# Local CLI is verification only
show managers
show conn
show version
```

Verify

```
FMC deployment status
Connection Events filtered by DENY-ICMPV4
FTD: show conn count
```

Check Point Quantum

MANAGEMENT API + SECUREXL DOS

Create policy on the management server, publish it, then install it to the gateway. SecureXL rate limiting is enabled on the gateway.

Block IPv4 ICMP transit

```
mgmt_cli add access-rule layer "LAB-Network" position 3 \
  name "DENY-ICMPV4" source "LAB-CLIENTS" \
  destination "LAB-PUBLIC-WEB" service "icmp-proto" \
  action "Drop" track.type "Log" -r true
mgmt_cli publish -r true
mgmt_cli install-policy policy-package "LAB-POLICY" \
  targets "LAB-GATEWAY" -r true
```

Enable flood / DDoS protection

```
# Expert mode on the Security Gateway
fwaccel dos config set --enable-rate-limit
fwaccel dos config set --enable-pbox
fwaccel dos config set --enable-log-drops
fwaccel dos rate add destination cidr:198.51.100.20/32 \
  service any pkt-rate 4000
fwaccel dos config get
fwaccel dos rate get
```

Verify

```
mgmt_cli show access-rulebase name "LAB-Network" -r true
fwaccel dos rate get
```

Platform commands: Sophos and Juniper

Sophos Firewall WEBADMIN + VERIFIED DOS-CONFIG CLI Create the traffic rule in WebAdmin. Sophos 21.5 documents CLI activation for SYN, UDP, and ICMP flood policies and rules. Block IPv4 ICMP transit <pre>WebAdmin > Rules and policies > Firewall rules 1. Add DENY-ICMPV4 2. Source zone LAN; destination zone WAN 3. Service: ICMP 4. Action: Drop; enable logging 5. Place below DNS/WEB allows</pre> Enable flood / DDoS protection <pre>system dos-config add dos-policy policy-name LAB-EDGE-DOS \ SYN-Flood 2000 pps per-dst \ UDP-Flood 8000 pps per-dst \ ICMP-Flood 300 pps per-dst system dos-config add dos-rule rule-name LAB-WAN-PROTECT \ src-zone WAN dstip 198.51.100.20 \ dos-policy LAB-EDGE-DOS</pre> Verify <pre>system dos-config show dos-policies policy-name LAB-EDGE-DOS system dos-config show dos-rules rule-name LAB-WAN-PROTECT</pre>	Juniper SRX JUNOS CLI IDS SCREENS A zone policy denies IPv4 ICMP. An IDS screen enables SYN, UDP, and ICMP flood controls on the untrust zone. Block IPv4 ICMP transit <pre>set security policies from-zone LAB-TRUST to-zone LAB-UNTRUST \ policy DENY-ICMPV4 match source-address any set security policies from-zone LAB-TRUST to-zone LAB-UNTRUST \ policy DENY-ICMPV4 match destination-address any set security policies from-zone LAB-TRUST to-zone LAB-UNTRUST \ policy DENY-ICMPV4 match application junos-icmp-all set security policies from-zone LAB-TRUST to-zone LAB-UNTRUST \ policy DENY-ICMPV4 then deny</pre> Enable flood / DDoS protection <pre>set security screen ids-option LAB-EDGE tcp syn-flood alarm-threshold 1200 set security screen ids-option LAB-EDGE tcp syn-flood attack-threshold 2000 set security screen ids-option LAB-EDGE tcp syn-flood source-threshold 500 set security screen ids-option LAB-EDGE tcp syn-flood destination-threshold 2000 set security screen ids-option LAB-EDGE tcp syn-flood timeout 20 set security screen ids-option LAB-EDGE udp flood threshold 8000 set security screen ids-option LAB-EDGE icmp flood threshold 300 set security zones security-zone LAB-UNTRUST screen LAB-EDGE commit check commit confirmed 5</pre> Verify <pre>show security screen ids-option LAB-EDGE show security screen statistics zone LAB-UNTRUST</pre>
--	---

Platform workflows: SonicWall and WatchGuard

SonicWall SONICOS MANAGER / CLI TREE Use Access Rules for the ICMP deny. Flood command grammar varies across SonicOS 7/8 and hardware families, so confirm each leaf with inline help. Block IPv4 ICMP transit <pre>SonicOS Manager > Policy > Rules and Policies 1. Add rule: DENY-ICMPV4 2. From LAN to WAN 3. Service: Ping / ICMP 4. Action: Deny; logging on 5. Place below DNS/WEB allows</pre> Enable flood / DDoS protection <pre># Version-dependent CLI pattern configure udp flood-protection flood-attack-threshold 8000 flood-block-timeout 30 commit best-effort exit # Also enable in UI Network > Firewall > Flood Protection - Layer 3 SYN Flood Protection - ICMP Flood Protection - WAN DDoS Protection</pre> Verify <pre>SonicOS Manager flood counters Access Rule hits for DENY-ICMPV4 CLI: show current-config match flood</pre>	WatchGuard Firebox FIREWARE CLI DEFAULT PACKET HANDLING Unhandled traffic is denied by default. Add an explicit ICMP deny when the lab needs visible rule-level evidence, then enable flood and DDoS quotas. Block IPv4 ICMP transit <pre>Policy Manager / Fireware Web UI 1. Add DENY-ICMPV4 policy 2. From Any-Trusted to Any-External 3. Packet filter: ICMP 4. Denied; logging enabled 5. Keep management/VPN policies separate</pre> Enable flood / DDoS protection <pre>default-packet-handling dangerous-activity syn-flood enable 2000 default-packet-handling dangerous-activity udp-flood enable 8000 default-packet-handling dangerous-activity icmp-flood enable 300 default-packet-handling ddos server-ddos enable 1500 default-packet-handling ddos client-ddos enable 1500</pre> Verify <pre>show default-packet-handling Traffic Monitor: filter deny reason / flood type</pre>
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Platform workflows: Forcepoint and Huawei

Forcepoint NGFW

NGFW MANAGER / SMC | MANAGER-DRIVEN

Access policy and DoS protection are installed from NGFW Manager. The engine CLI is for status and troubleshooting, not invented policy authoring.

Block IPv4 ICMP transit

```
NGFW Manager > Access Policy
1. Allow DNS and WEB rules
2. Add DENY-ICMPV4 below the allows
3. Match LAB-CLIENTS -> LAB-UNTRUST
4. Service: ICMP; action: Discard + Log
5. Validate and install policy
```

Enable flood / DDoS protection

```
NGFW Manager > Engine Editor > Advanced Settings
1. Enable Rate-Based DoS Protection Mode
2. Enable SYN flood protection
3. Enable slow HTTP protection where relevant
4. Set measured thresholds
5. Install engine configuration and policy
```

Verify

```
NGFW Manager policy validation
Engine status / alerts
Log Browser: DENY-ICMPV4 and DoS events
```

Huawei HiSecEngine

VRP-STYLE CLI | ANTI-DDOS

A security policy denies ICMP transit. Flow statistics and detect-clean mode activate the anti-DDoS workflow for the lab WAN.

Block IPv4 ICMP transit

```
system-view
security-policy
rule name DENY-ICMPV4
source-zone LAB-TRUST
destination-zone LAB-UNTRUST
service protocol icmp
action deny
quit
save
```

Enable flood / DDoS protection

```
system-view
interface GigabitEthernet0/0/1
anti-ddos flow-statistic enable
quit
ddos-mode detect-clean
anti-ddos ip user-defined enable
# Confirm release-specific SYN/UDP/ICMP leaves with ?
save
```

Verify

```
display security-policy rule all
display current-configuration | include anti-ddos
display current-configuration interface GigabitEthernet0/0/1
```


Validation and rollback runbook

A recruiter-facing firewall lab should prove the control state, the allowed business traffic, the denied traffic, and the recovery path. The test plan below stays intentionally small enough to execute and explain clearly.

Test	Fake target	Expected result	Evidence
DNS UDP	203.0.113.53:53	Allowed	Query succeeds; allow-rule counter increments.
DNS TCP	203.0.113.53:53	Allowed	TCP fallback succeeds; session closes normally.
HTTP	198.51.100.20:80	Allowed	HTTP response header; allow-rule counter increments.
HTTPS	198.51.100.20:443	Allowed	TLS session and HTTP response; logged at session end.
IPv4 ping	198.51.100.20	Blocked	No echo reply; DENY-ICMPV4 match is logged.
SSH	198.51.100.20:22	Blocked	Connection fails; cleanup deny counter increments.
Controlled SYN/UDP/ICMP rate test	198.51.100.20	Protection triggers at lab threshold	Alarm/drop counter rises; permitted DNS/web remains healthy.

Fake endpoint commands

```
# Authorized documentation-only targets
nslookup portal.example 203.0.113.53
curl -I http://198.51.100.20/
curl -I https://198.51.100.20/
ping -4 198.51.100.20           # expected: blocked
nc -vz 198.51.100.20 22        # expected: blocked

# Flood testing requires an isolated lab traffic generator.
# Never send high-rate traffic to public systems or shared networks.
```

BEFORE Export current policy, verify console/OOB access, record rule order, capture counters, and confirm management reachability.	DURING Apply one bounded change, watch CPU/session health, test allow and deny paths, and confirm logs are rate-limited.	ROLLBACK Revert the policy/profile, commit or deploy, remove temporary test objects, and re-run the known-good checks.	AFTER Save sanitized evidence: rule IDs, counters, timestamps, firmware version, finding, and final disposition.
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What this project demonstrates

The playbook keeps the configuration goal simple while showing the habits required for larger firewall programs: clear scope, least privilege, platform-aware implementation, measurable availability controls, rollback safety, and evidence-based validation.

Capability	Portfolio evidence
Policy design	Four explicit allowed transports, logged IPv4 ICMP denial, cleanup deny, correct rule order, and stateful return traffic.
DDoS defense	Visible activation of SYN, UDP, ICMP, and connection-rate protections, plus the upstream-saturation boundary.
Multi-vendor judgment	CLI is used where supported; manager workflows replace fabricated commands where persistent policy is manager-driven.
Public safety	RFC-reserved addresses, .example names, fake objects, illustrative thresholds, and no production identifiers or credentials.
Operational discipline	Baseline, monitor, protect, test, tune, document, and roll back as a repeatable change sequence.
Technical honesty	Commands are version-bound patterns and require exact release validation before any production use.

Official documentation starting points

Platform	Primary reference
Palo Alto	docs.paloaltonetworks.com - Zone Protection Profiles and Flood Protection
Fortinet	docs.fortinet.com - FortiOS firewall and DoS policy administration
Cisco	cisco.com - Secure Firewall Access Control and Threat Defense Service Policy
Check Point	sc1.checkpoint.com - Management API and SecureXL fwaccel dos
Sophos	docs.sophos.com - Sophos Firewall 21.5 system dos-config
Juniper	juniper.net/documentation - SRX IDS screens and screen statistics
SonicWall	sonicwall.com/support - SonicOS Access Rules and Flood Protection
WatchGuard	watchguard.com/help - Default Packet Handling and DDoS quotas
Forcepoint	help.forcepoint.com/ngfw - Access Policy and Rate-Based DoS
Huawei	support.huawei.com - HiSecEngine USG security policy and DDoS defense

RELEASE VALIDATION REQUIRED

Firewall command trees, manager APIs, object names, and feature licenses change. Compare every pattern with the exact model, firmware release, management mode, and deployment target before use.

PORTFOLIO CONCLUSION

This document is a public-safe engineering reference, not proof that ten physical firewalls were deployed. Runtime screenshots, counters, and sanitized packet evidence should be added only after authorized lab execution.